Harsh Singh

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Professional Summary

Computer Science undergraduate specializing in Cyber Security with expertise in Java, Python, and JavaScript. Hands-on experience in full-stack development, cloud infrastructure, and distributed systems. **AWS Certified Solutions Architect** with proven track record of deploying enterprise automation solutions that reduced operational overhead by 40%. Strong foundation in Data Structures & Algorithms, Object-Oriented Programming, and CI/CD pipelines. Seeking Software Development Engineer internship to build scalable, production-grade systems.

Education

Noida Institute of Engineering & Technology — Greater Noida, Uttar Pradesh
Bachelor of Technology in Computer Science Engineering (Cyber Security); CGPA: 7.5/10.0 (75%) — *Expected June 2026*

Kendriya Vidyalaya (AFS) Uttarlai — Barmer, Rajasthan
Class 12th, CBSE Board; 86% Overall (95% in Mathematics) — *2022*

Technical Skills

Programming Languages: Java, Python, JavaScript, SQL, Bash, HTML5, CSS3, Google Apps Script
Cloud & DevOps: Amazon Web Services (AWS), EC2, S3, Lambda, IAM, SNS, CloudWatch, Terraform, Infrastructure as Code (IaC), CI/CD Pipelines, GitHub Actions, Blue-Green Deployment, Docker
Web Technologies: React.js, Node.js, Express.js, RESTful APIs, Responsive Design, Single Page Applications (SPA), Cross-Browser Compatibility
Databases: MongoDB, MySQL, NoSQL, SQL Query Optimization
Tools & Platforms: Git, GitHub, Postman, Linux (Ubuntu, Kali), Vercel, VS Code, npm
Core Concepts: Data Structures & Algorithms, Object-Oriented Programming (OOP), System Design, Distributed Systems, Software Development Life Cycle (SDLC), Agile Methodologies, Scalability, Fault Tolerance, Security Best Practices

Professional Experience

Cloud Infrastructure Intern — July 2025 – December 2025
MERITTO (NoPaperForms) — *Noida, India*

- Administered and optimized AWS cloud infrastructure across multiple services including EC2 instance management, S3 bucket configuration, Lambda function deployment, and IAM policy implementation, ensuring secure and scalable infrastructure
- Engineered and deployed 'MAIL BOT', an enterprise automation solution using Google Apps Script and OpenAI API, processing 500+ automated communications monthly and reducing manual operational overhead by **40%**
- Designed end-to-end communication pipeline integrating unified OpenAPI platform with 15+ customized email templates, automating Minutes of Meeting (MoM) lifecycle from generation to distribution across stakeholder teams
- Implemented CI/CD best practices and blue-green deployment strategies, utilizing GitHub Secrets Manager for secure credential management and automated infrastructure deployment
- Collaborated with cross-functional teams following Agile methodologies to deliver cloud solutions aligned with business requirements and security compliance standards

Web Development Intern — August 2024 – August 2024
Codsoft — *Remote*

- Developed responsive web applications using HTML5, CSS3, and JavaScript, implementing mobile-first design principles and ensuring cross-browser compatibility across Chrome, Firefox, Safari, and Edge
- Created professional portfolio website and landing page demonstrations showcasing modern UI/UX design patterns, responsive layouts, and optimized performance metrics
- Implemented accessibility standards (WCAG 2.1) and performance optimization techniques achieving 90+ Lighthouse scores across all web vitals categories (Performance, Accessibility, Best Practices, SEO)

Technical Projects

AWS Cost AI Optimizer | *Terraform, AWS Lambda, Python, Scikit-Learn, S3, SNS* — GitHub

- Architected serverless MLOps pipeline using Infrastructure as Code (Terraform) and AWS Lambda to automate billing data ingestion into S3 Data Lake, transforming reactive cost management into predictive analytics framework

- Built machine learning forecasting engine using Python and Scikit-Learn to analyze 6 months of historical spending patterns and predict end-of-month costs with 85% accuracy, triggering proactive Amazon SNS alerts when budget thresholds exceeded

- Implemented AWS security best practices including IAM Least Privilege policies, encryption at rest and in transit, and environment-driven configuration management for production-grade deployment

AI-Enabled Terminal Portfolio | *React.js, Node.js, OpenAI API, Vercel* Live Demo

- Developed interactive terminal-inspired portfolio leveraging React.js component architecture with command-line navigation interface, demonstrating advanced JavaScript programming and state management capabilities
- Integrated custom AI chatbot widget powered by OpenAI API to deliver real-time intelligent responses, enhancing user engagement and providing dynamic interaction capabilities
- Deployed scalable application on Vercel cloud platform with automated CI/CD pipeline ensuring zero-downtime deployments and optimal performance

AI MAIL BOT | *Google Apps Script, HTML/CSS, JavaScript, OpenAI API* GitHub

- Developed full-stack automation platform combining Google Apps Script backend with OpenAI API integration to transform raw meeting transcripts into professional, template-driven client communications for Operations teams
- Built custom WYSIWYG editor with responsive frontend enabling seamless synthesis of AI-generated content with 20+ brand-specific email templates, ensuring 100% consistency in client-facing communications
- Automated technical data extraction including ticket numbers, URLs, and metadata validation, eliminating human error and reducing email drafting time by 60%

Certifications & Professional Development

Cloud & Infrastructure: AWS Certified Solutions Architect – Associate, AWS Academy Cloud Foundations, IBM Linux Commands & Shell Scripting, Cisco Operating System Basics
Programming & Development: HackerRank Python Programming Certification, Apple App Development with Swift (Level 1)
Cyber Security: EC-Council SQL Injection Attacks Certification, Security Best Practices

Honors & Achievements

Top 20 Finalist – Smart India Hackathon 2023: National-level competition focused on Blockchain-based Certificate Authentication system
First Place Winner – Inter-College Design Thinking Competition: Led team to victory in innovation-focused design challenge
Speaker – Phishing Attacks Workshop: Delivered technical presentation on cybersecurity threats to 100+ students
Participant – Segue 2.0 Global Design Thinking Challenge: Competed in international innovation and problem-solving competition